Industrial Vision

Lecture 12 - Camera calibration

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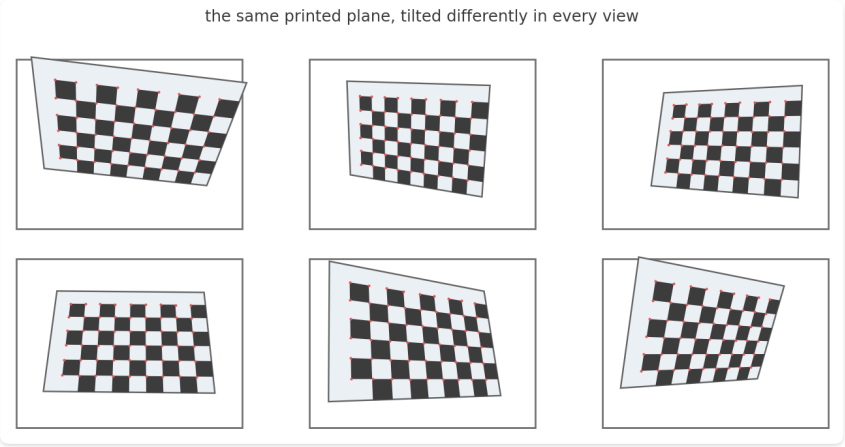
Today we obtain K. Several photographs of a printed plane, in practice, is used to estimate K by **Zhang's method**. A single photograph of a scene containing three mutually orthogonal sets of parallel lines suffices in principle, through the **vanishing points** and the **image of the absolute conic**. We close with the one effect no 3×3 matrix can represent, namely **lens distortion**.

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Calibration using a planar target

The instrument used everywhere in practice is a **printed checkerboard**, photographed several times in different orientations. It has three virtues.

- Its corners are detectable to **sub-pixel accuracy** by the methods of **L03** and **L05**, and sub-pixel accuracy is what makes calibration accurate.
- Their world coordinates are **known exactly** by construction, since the squares were printed at a known size.
- The target is **planar**, so the homography of **L13** applies to it directly, and that same lecture's estimator applies to the homography.



Zhang: the homography of a plane, and what it constrains

Place the target's plane at $Z = 0$ in the world frame, so that a target point is $(X, Y, 0, 1)^\top$. Then

$$s\,\tilde{\mathbf{x}} = K[\,\mathbf{r}_1\ \mathbf{r}_2\ \mathbf{r}_3\ \mathbf{t}\,]\begin{bmatrix} X \\ Y \\ 0 \\ 1 \end{bmatrix} = K[\,\mathbf{r}_1\ \mathbf{r}_2\ \mathbf{t}\,]\begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

The third column of R is annihilated by the zero. What survives is a 3×3 matrix acting on a planar point, which is to say a **homography**:

$$\boxed{H = \lambda\, K[\,\mathbf{r}_1\ \mathbf{r}_2\ \mathbf{t}\,]} \qquad H = [\,\mathbf{h}_1\ \mathbf{h}_2\ \mathbf{h}_3\,]$$

with λ an unknown scale, since H is projective and is only ever determined up to scale.

This H is obtained by the normalized DLT of L13, from the known target corners and their detected images.

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Two constraints per view

- The board is photographed in n views and the DLT is run **once per view**. Write H^k for the homography of view k and $\mathbf{h}_i^k$ for its i-th column:

$$H^k = \begin{bmatrix} \mathbf{h}_1^k & \mathbf{h}_2^k & \mathbf{h}_3^k \end{bmatrix} = \begin{bmatrix} h_{11}^k & h_{21}^k & h_{31}^k \\ h_{12}^k & h_{22}^k & h_{32}^k \\ h_{13}^k & h_{23}^k & h_{33}^k \end{bmatrix}, \qquad \mathbf{h}_i^k = \begin{bmatrix} h_{i1}^k \\ h_{i2}^k \\ h_{i3}^k \end{bmatrix}, \qquad k = 1, \dots, n$$

- The second index runs **down** a column, so h_{i2}^k is the second entry of the i-th column of view k's homography.

- Each view carries its own scale λ_k and its own pose. Inverting $H^k = \lambda_k K \begin{bmatrix} \mathbf{r}_1^k & \mathbf{r}_2^k & \mathbf{t}^k \end{bmatrix}$ column by column, $\mathbf{r}_1^k = \lambda_k^{-1} K^{-1} \mathbf{h}_1^k$ and $\mathbf{r}_2^k = \lambda_k^{-1} K^{-1} \mathbf{h}_2^k$. By **L12** these are columns of a **rotation matrix**, so they are orthogonal and of equal, unit, norm. Two facts, two equations, **for every** k:

$$\mathbf{r}_1^{k\top} \mathbf{r}_2^k = 0 \implies \mathbf{h}_1^{k\top} K^{-\top} K^{-1} \mathbf{h}_2^k = 0, \qquad \|\mathbf{r}_1^k\| = \|\mathbf{r}_2^k\| \implies \mathbf{h}_1^{k\top} K^{-\top} K^{-1} \mathbf{h}_1^k = \mathbf{h}_2^{k\top} K^{-\top} K^{-1} \mathbf{h}_2^k$$

- The unknown λ_k cancels in both, which is why a ratio and a difference are used rather than $\|\mathbf{r}_1^k\| = 1$. Across the n views this is $2n$ equations in one and the same unknown matrix $K^{-\top} K^{-1}$.

$$\mathbf{h}_1^{k\top} B \mathbf{h}_2^k = 0$$

$$\mathbf{h}_1^{k\top} B \mathbf{h}_1^k = \mathbf{h}_2^{k\top} B \mathbf{h}_2^k$$

Writing $\mathbf{h}_i^\top B\,\mathbf{h}_j$ out

Fix one view and drop the superscript k until the last line. Both constraints have the same shape, $\mathbf{h}_i^\top B\,\mathbf{h}_j$, with B **symmetric**, so it holds only six distinct numbers:

$$B \;=\; K^{-\top}K^{-1} \;=\; \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{12} & B_{22} & B_{23} \\ B_{13} & B_{23} & B_{33} \end{bmatrix}$$

Multiply the row, the matrix and the column out, and gather the nine terms by those six entries, using $B_{pq} = B_{qp}$:

$$\begin{bmatrix} h_{i1} & h_{i2} & h_{i3} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} & B_{13} \\ B_{12} & B_{22} & B_{23} \\ B_{13} & B_{23} & B_{33} \end{bmatrix} \begin{bmatrix} h_{j1} \\ h_{j2} \\ h_{j3} \end{bmatrix} \;=\; \begin{aligned} &B_{11}\,h_{i1}h_{j1} \;+\; B_{12}\big(h_{i1}h_{j2} + h_{i2}h_{j1}\big) \;+\; B_{22}\,h_{i2}h_{j2} \\ &+\; B_{13}\big(h_{i1}h_{j3} + h_{i3}h_{j1}\big) \;+\; B_{23}\big(h_{i2}h_{j3} + h_{i3}h_{j2}\big) \;+\; B_{33}\,h_{i3}h_{j3} \end{aligned}$$

The right-hand side is a **sum of six products**, each a known number times an unknown entry of B: it is **linear in** B. Put the unknowns in one vector and the known products in another,

$$\mathbf{b} = \begin{bmatrix} B_{11} \\ B_{12} \\ B_{22} \\ B_{13} \\ B_{23} \\ B_{33} \end{bmatrix}, \qquad \mathbf{v}_{ij} = \begin{bmatrix} h_{i1}h_{j1} \\ h_{i1}h_{j2} + h_{i2}h_{j1} \\ h_{i2}h_{j2} \\ h_{i1}h_{j3} + h_{i3}h_{j1} \\ h_{i2}h_{j3} + h_{i3}h_{j2} \\ h_{i3}h_{j3} \end{bmatrix}, \qquad \text{so that} \qquad \boxed{\mathbf{h}_i^\top B\,\mathbf{h}_j = \mathbf{v}_{ij}^\top \mathbf{b}}$$

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$$\boxed{\mathbf{h}_i^\top B\,\mathbf{h}_j = \mathbf{v}_{ij}^\top \mathbf{b}}$$

$$\mathbf{h}_1^\top B \mathbf{h}_2 = \mathbf{v}_{12}^\top \mathbf{b} = 0$$

$$\mathbf{h}_1^\top B \mathbf{h}_1 = \mathbf{h}_2^\top B \mathbf{h}_2 \Longrightarrow \mathbf{v}_{11}^\top \mathbf{b} = \mathbf{v}_{22}^\top \mathbf{b} \Longrightarrow (\mathbf{v_{11}} - \mathbf{v_{22}})^\top \mathbf{b} = 0$$

Every entry of $\mathbf{v}_{ij}$ is a number read off the estimated H; the only unknown left anywhere is $\mathbf{b}$. As a check, $i = j = 1$ gives $\mathbf{v}_{11} = \big[\, h_{11}^2,\ 2h_{11}h_{12},\ h_{12}^2,\ 2h_{11}h_{13},\ 2h_{12}h_{13},\ h_{13}^2 \,\big]^\top$.

Restore the view index. The two constraints of view k are $\mathbf{v}_{12}^{k\top}\mathbf{b} = 0$ and $\mathbf{v}_{11}^{k\top}\mathbf{b} = \mathbf{v}_{22}^{k\top}\mathbf{b}$, where $\mathbf{v}_{ij}^k$ is built from the columns of H^k alone. That is **two linear rows per view**, in the same six unknowns for all views:

$$V^k\mathbf{b} \;=\; \begin{bmatrix} \mathbf{v}_{12}^{k\top} \\ \big(\mathbf{v}_{11}^k - \mathbf{v}_{22}^k\big)^\top \end{bmatrix} \mathbf{b} \;=\; \mathbf{0}, \qquad V^k \in \mathbb{R}^{2\times 6}, \qquad k = 1, \dots, n$$

The full system: n views, $2n$ rows, six unknowns

The n views are processed independently as far as their own two rows, then **stacked**, in any order:

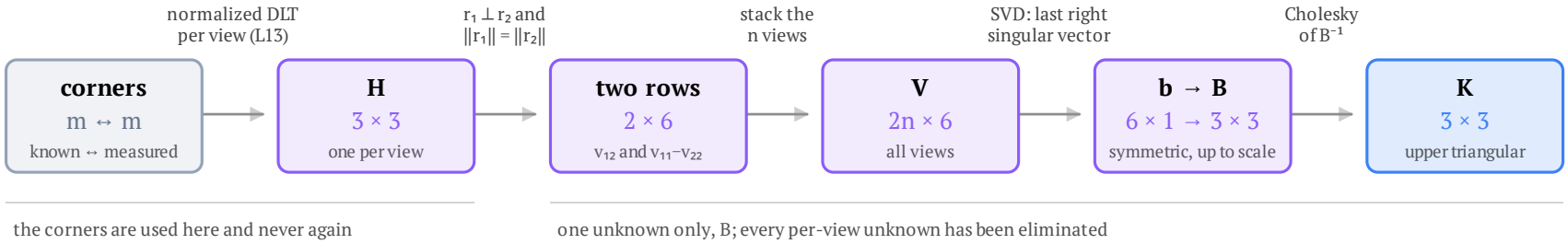
$$\underbrace{\begin{bmatrix} V^{(1)} \\ V^{(2)} \\ \vdots \\ V^{(n)} \end{bmatrix}}_{V} \mathbf{b} = \underbrace{\begin{bmatrix} \mathbf{v}_{12}^{(1)\top} \\ \left(\mathbf{v}_{11}^{(1)} - \mathbf{v}_{22}^{(1)}\right)^{\top} \\ \vdots \\ \mathbf{v}_{12}^{(n)\top} \\ \left(\mathbf{v}_{11}^{(n)} - \mathbf{v}_{22}^{(n)}\right)^{\top} \end{bmatrix}}_{2n \times 6} \underbrace{\begin{bmatrix} B_{11} \\ B_{12} \\ B_{22} \\ B_{13} \\ B_{23} \\ B_{33} \end{bmatrix}}_{6 \times 1} = \underbrace{\mathbf{0}}_{2n \times 1} \quad \Longleftrightarrow \quad \boxed{V\mathbf{b} = \mathbf{0}, \quad V \in \mathbb{R}^{2n \times 6}}$$

- **This is the equation of Zhang's linear step**, and it is the whole of it. Two rows enter per view, so V grows downwards as views are added while its six columns never change.
- $\mathbf{b}$ is fixed only **up to scale**, so five independent rows suffice, and two rows per view gives $2n \geq 5$, hence $\mathbf{n \geq 3}$ views. Rank, not row count, is what matters: rows from views of the same attitude are dependent and add nothing.
- Every entry of V is a **product of two entries of an estimated homography**, a known number. The per-view unknowns λ_k, $\mathbf{r}_3^k$ and $\mathbf{t}^k$ appear nowhere in it, having been eliminated rather than solved for.

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From corners to K: what is on the table at each stage

One board of m inner corners, photographed in n views. Read left to right: the boxes are the data, the labels on the arrows are the operations.

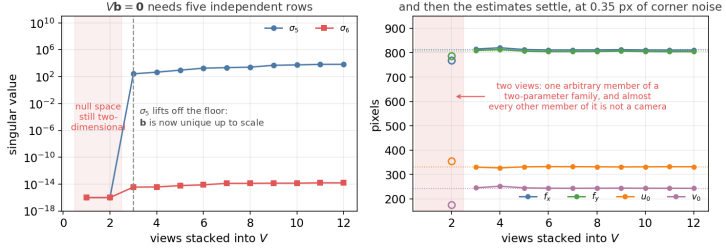


- The corners enter **exactly once**, at the DLT. A 9 × 6 board's fifty-four corners are compressed into the nine numbers of H, and every later step sees only those nine. More corners help by making H better, not by adding equations to V.
- λ_i, $\mathbf{r}_3$ and $\mathbf{t}_i$ are never solved for here, they are **eliminated**. That is exactly why the two chosen facts are a perpendicularity and a ratio: λ cancels in both.
- B is a **change of variable**, nothing more. $\mathbf{h}_i^\top B\, \mathbf{h}_j$ is linear in the six entries of B but badly nonlinear in f_x, f_y, s, p_x, p_y; the Cholesky at the end undoes the substitution.

Solving $V\mathbf{b} = \mathbf{0}$, and recovering K

With $V \in \mathbb{R}^{2n\times 6}$ built, solve $V\mathbf{b} = \mathbf{0}$ by the **last right singular vector of** V, which is identically the machinery of **L13** on a different matrix.

- Three views are the minimum. With zero skew assumed one may manage with $n = 2$, and in practice ten to twenty views are used.
- The views must differ in **orientation**, not merely in position, or the new rows duplicate the old ones.
- Since $B = K^{-\top}K^{-1}$ we have $B^{-1} = K\,K^{\top}$, so **factorizing** B^{-1} **into an upper-triangular factor with positive diagonal**, the Cholesky factorization, returns K directly. Zhang gives equivalent closed forms for f_x, f_y, s, p_x, p_y, which implementations usually use.
- B must be **positive definite** for this to succeed and noise can violate it, which is one reason the linear solution is only an initializer.



The per-view pose, and putting R back on $SO(3)$

With K known, return to $H^k = \lambda_k K \begin{bmatrix} \mathbf{r}_1^k & \mathbf{r}_2^k & \mathbf{t}^k \end{bmatrix}$ and read off the pose of each of the n views. Writing $\mu_k = \lambda_k^{-1}$, the scale is fixed by requiring $\|\mathbf{r}_1^k\| = 1$:

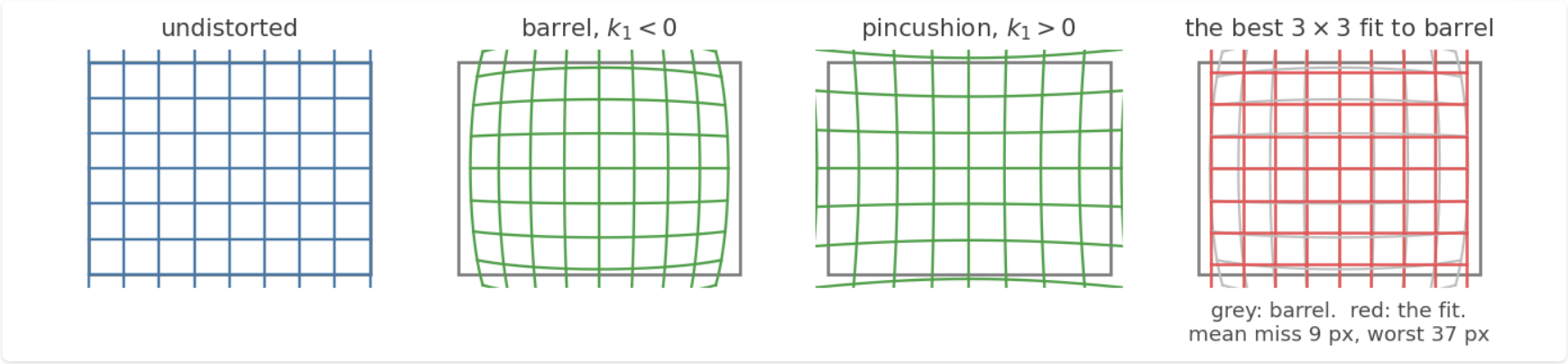
$$\mu_k = \frac{1}{\|K^{-1}\mathbf{h}_1^k\|}, \qquad \mathbf{r}_1^k = \mu_k K^{-1}\mathbf{h}_1^k, \quad \mathbf{r}_2^k = \mu_k K^{-1}\mathbf{h}_2^k, \quad \mathbf{r}_3^k = \mathbf{r}_1^k \times \mathbf{r}_2^k, \quad \mathbf{t}^k = \mu_k K^{-1}\mathbf{h}_3^k$$

The third column is recovered by a **cross product**, because it must complete a right-handed orthonormal triple.

The estimated $R^k = \begin{bmatrix} \mathbf{r}_1^k & \mathbf{r}_2^k & \mathbf{r}_3^k \end{bmatrix}$ is not exactly a rotation, because $\mathbf{h}_1^k$ and $\mathbf{h}_2^k$ carry noise. Project it onto $SO(3)$: take the singular value decomposition $R^k = U\Sigma V^\top$

$$\boxed{\hat{R}^k = UV^\top}$$

Lens distortion, and why it is not in K



Real lenses bend straight lines. Model the effect in **normalized** coordinates $(x_n, y_n) = (X_c/Z_c,\ Y_c/Z_c)$, the ideal pinhole image on the plane $Z = 1$, before K converts it to pixels, and let $r^2 = x_n^2 + y_n^2$:

$$x_d = x_n\left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6\right) + \underbrace{2p_1 x_n y_n + p_2\left(r^2 + 2x_n^2\right)}_{\text{tangential}}, \qquad y_d = y_n\left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6\right) + \underbrace{p_1\left(r^2 + 2y_n^2\right) + 2p_2 x_n y_n}_{\text{tangential}}$$

Only then is K applied, to reach pixels. The radial terms are even powers of r by symmetry, giving **barrel** distortion when $k_1 < 0$ and **pincushion** when $k_1 > 0$. The tangential terms arise when lens and sensor are not parallel, and are usually small.

The full calibration procedure

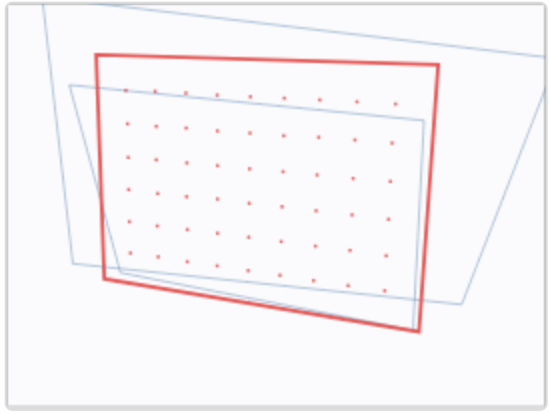
1. Detect the target corners to **sub-pixel accuracy** (L03, L05), in the raw, **distorted** image.
2. For each view, estimate H by the **normalized DLT** (L13).
3. Build V, solve $V\mathbf{b} = \mathbf{0}$ by SVD, and recover K by Cholesky of B^{-1}.
4. Recover $(R_i, \mathbf{t}_i)$ for each view, and project each R_i onto $SO(3)$. Steps 2–4 pretend the lens is ideal.
5. Initialize $\mathbf{k} = (k_1, k_2, p_1, p_2, k_3) = \mathbf{0}$.
6. **Refine everything jointly**, meaning K, all the poses and all the distortion coefficients, by minimizing the total reprojection error

$$\min \sum_i \sum_j \left\| \mathbf{x}_{ij} - \pi\left(K,\, \mathbf{k},\, R_i,\, \mathbf{t}_i,\, \mathbf{X}_j\right) \right\|^2$$

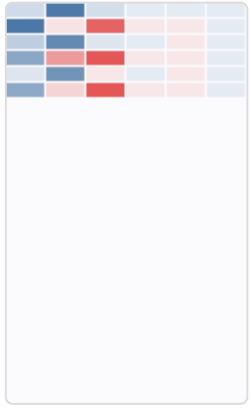
where π maps $\mathbf{X}_j$ by $(R_i, \mathbf{t}_i)$ into the camera frame, divides by depth to reach (x_n, y_n), **distorts** by $\mathbf{k}$ with the polynomial of the previous slide, and only then applies K. Distortion is applied forward here, so the prediction is compared with the distorted corner as detected.

Demonstration of calibration

A · Zhang, view by view | B · distortion vs a 3×3 | C · vanishing points



the first 3 views of the target



V, 6 × 6

views 3/12

6 rows, of the 5 independent equations needed: **$\sigma_5/\sigma_6 \approx 10^5$, so b is unique up to scale**

	recovered	true
f_x	814.8	812.0
f_y	809.2	806.0
u_0	330.1	331.0
v_0	245.1	243.0

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Live: calibrating this webcam

Three things to watch.

- After the first and second capture, σ_5 of V sits on the numerical floor and B is not positive definite.
- Translate the board without tilting it and capture again: σ_5 does not move. Orientation, not position, adds rank.
- Two answers appear. The **linear** K is the SVD and Cholesky of the slides; the **refined** K adds the five distortion coefficients and the nonlinear step deferred to **L16**. Their focal lengths differ by tens of pixels.

key	action
`SPACE`	capture the view, if the board is found
`d`	drop the last view
`u`	toggle the undistorted feed
`s`	save K and $\mathbf{k}$ to disk

+ board

Another way to do the calibration (from a single image) – Points at infinity project to finite points

We know that a point of the form $(d_1, d_2, d_3, 0)$ denotes a **direction** and not a location. Project one through the camera:

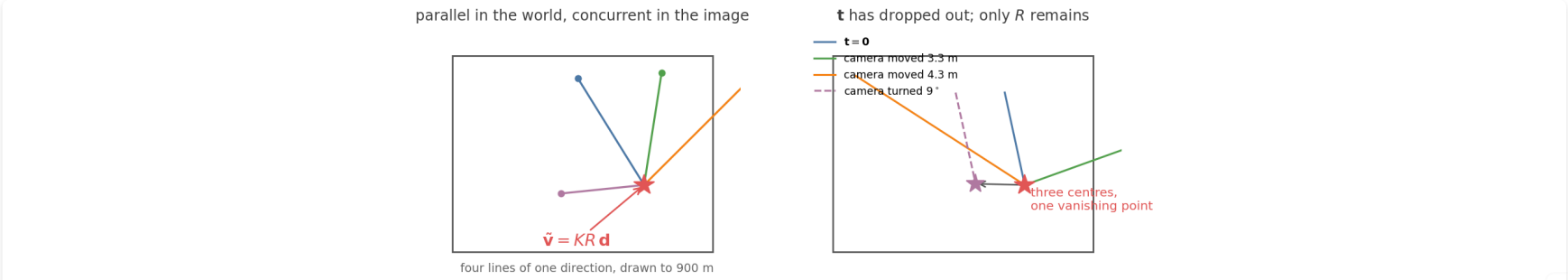
$$\tilde{\mathbf{v}} \;=\; P \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix} \;=\; K[\,R \mid \mathbf{t}\,] \begin{bmatrix} \mathbf{d} \\ 0 \end{bmatrix} \;=\; K R\,\mathbf{d}$$

The translation is multiplied by the zero and disappears: **the image of a direction does not depend on where the camera is, only on how it is turned.**

Writing $\mathbf{d}_c = R\mathbf{d}$ for that direction in the camera frame,

$$\tilde{\mathbf{v}} \sim K\,\mathbf{d}_c \qquad \Longleftrightarrow \qquad \mathbf{d}_c \sim K^{-1}\tilde{\mathbf{v}}$$

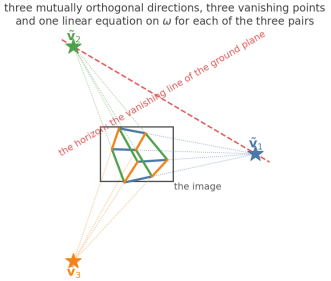
The second form does the work today: **an image point, back-projected through K^{-1}, is a direction in space.**



Vanishing points and the horizon

A **vanishing point** is the image of the point at infinity of a family of parallel lines. Every line with direction $\mathbf{d}$ passes through $\tilde{\mathbf{v}}$ in the image, however far apart those lines lie in the world: railway tracks, the edges of a corridor, the verticals of a building.

- The **vanishing line**, or horizon, of a plane is the image of that plane's line at infinity. Every direction lying in the plane has its vanishing point on that line, and for the ground plane it is the visible horizon.
- A vanishing point is **finite** unless the direction happens to be parallel to the image plane. In that case the parallel lines stay parallel in the image and the vanishing point goes off to infinity.



The image of the absolute conic

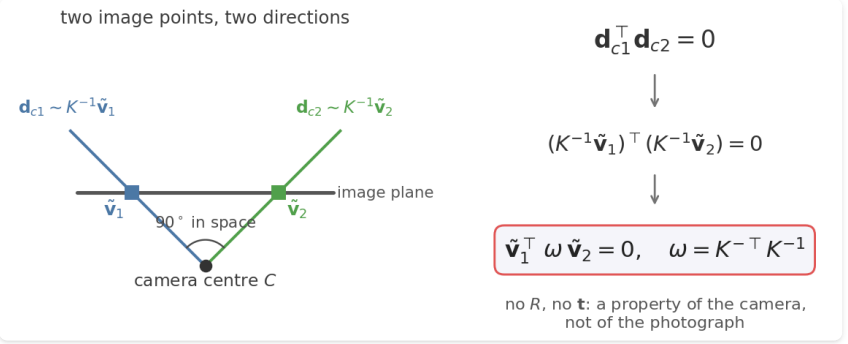
Take two directions $\mathbf{d}_1$ and $\mathbf{d}_2$ that are **orthogonal in the world**, with vanishing points $\tilde{\mathbf{v}}_1$ and $\tilde{\mathbf{v}}_2$. A rotation preserves inner products, so the two directions are still orthogonal in the camera frame:

$$\mathbf{d}_{c1}^{\top}\mathbf{d}_{c2} = 0$$

Substituting $\mathbf{d}_{ci} \sim K^{-1}\tilde{\mathbf{v}}_i$ from the previous slide,

$$\left(K^{-1}\tilde{\mathbf{v}}_1\right)^{\top}\left(K^{-1}\tilde{\mathbf{v}}_2\right) = 0 \implies \boxed{\tilde{\mathbf{v}}_1^{\top}\,\omega\,\tilde{\mathbf{v}}_2 = 0, \qquad \omega \;=\; K^{-\top}K^{-1}}$$

- The symmetric matrix ω is the **image of the absolute conic**. It depends on K **alone**, not on R and not on $\mathbf{t}$. It is a property of the **camera**, not of the photograph, and that is precisely why one photograph can calibrate.
- It is symmetric with six entries and is defined up to scale, so it carries **five degrees of freedom**, the same five as K, namely f_x, f_y, s, p_x and p_y.
- Each orthogonal pair of vanishing points therefore gives **one linear equation** in the six entries of ω.



Calibration from three orthogonal vanishing points

Three mutually orthogonal directions give three pairs, and therefore three equations. Three is fewer than five, so two assumptions must close the gap. The standard pair is **zero skew**, $s = 0$, and **square pixels**, $f_x = f_y$, which leaves the three unknowns f, u_0 and v_0 and makes the system exactly determined.

In practice. Photograph the corner of a room or a rectangular building. Fit three families of parallel lines with the **Hough transform**, intersect each family to obtain $\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \tilde{\mathbf{v}}_3$, and solve. The method works on found photographs, where no calibration target was ever available, and it is the only method in this course that calibrates from a **single image**.

However, it needs a scene that contains three orthogonal directions. It needs the two assumptions above. Worst of all, the vanishing points are **badly conditioned** when the lines are nearly parallel in the image, since a small angular error moves a distant intersection a very long way.

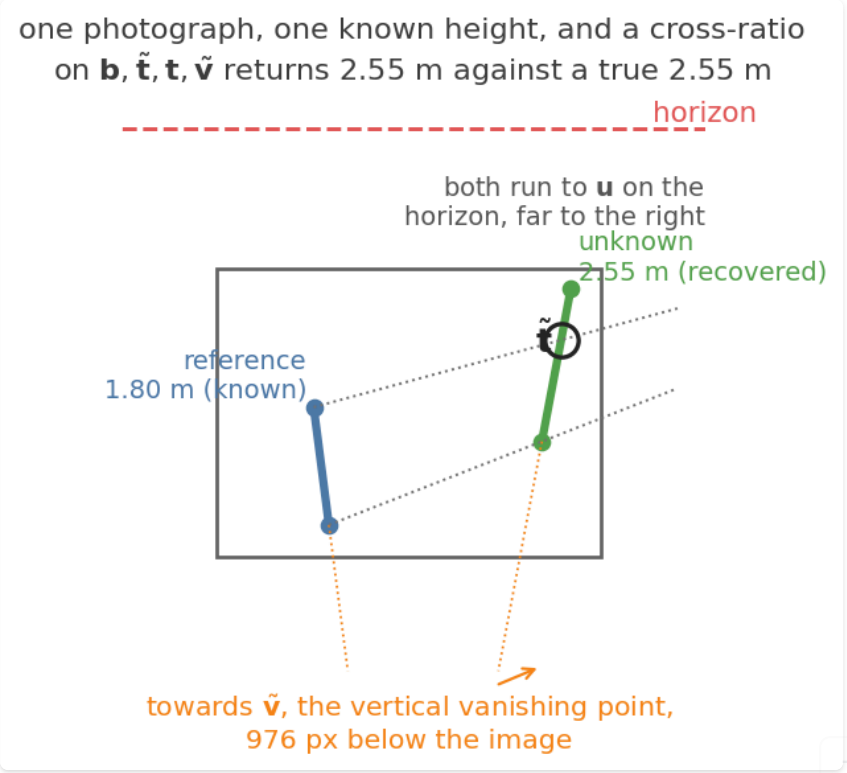
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Single-view metrology, briefly

Given the vanishing line of the ground plane and the vertical vanishing point, the height of an object **relative to a reference of known height** is recoverable from a single photograph.

- Join the two bases and extend that line to the horizon, giving a point $\mathbf{u}$.
- The line from $\mathbf{u}$ through the reference top cuts the unknown object's vertical at $\tilde{\mathbf{t}}$, which is the reference height **transferred** onto that vertical.
- The four points $\mathbf{b}, \tilde{\mathbf{t}}, \mathbf{t}, \tilde{\mathbf{v}}$ are collinear, and their **cross-ratio** is projectively invariant, so it returns the ratio of the two heights.

The technique is used in forensic photogrammetry to estimate the height of a person from surveillance footage. In the figure it returns 2.55 m against a true 2.55 m.



Questions

Q. Why must the calibration views differ in orientation rather than merely in position?

A. Because each view contributes rows built from $\mathbf{h}_1$ and $\mathbf{h}_2$, which encode the target's **attitude**. Translating the target parallel to itself changes $\mathbf{t}$, hence $\mathbf{h}_3$, but leaves $\mathbf{r}_1$ and $\mathbf{r}_2$ unchanged, so the two new rows duplicate the old ones and V gains no rank at all. This is why the board is tilted noticeably between shots.

Q. If distortion must be removed before anything else, why does the lecture estimate it last?

A. Because the linear stages provide the initial K and poses that the nonlinear stage needs. The pipeline is circular only in appearance: estimate the linear model pretending distortion is absent, then refine everything jointly with distortion switched on.

Q. Does the focal length recovered by calibration have units of millimetres?

A. It does not. f_x and f_y are in **pixels**, being the physical focal length divided by the pixel pitch in each direction. Recovering millimetres requires knowing the sensor's physical size, which no image-based method can supply. images determine geometry only up to a scale that must come from outside.

+ board

References

1. Zhang, Zhengyou. "A flexible new technique for camera calibration." IEEE Transactions on pattern analysis and machine intelligence 22.11 (2000): 1330-1334.
2. R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge Univ. Press, 2004. See **Ch. 8**, and in particular §8.5 on the absolute conic and its image. pringer, 2022. See §11.1 on calibration.

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